MAR ATHANASIUS COLLEGE OF ENGINEERING
KOTHAMANGALAM

(Government Aided Autonomous Institution)

Department of Computer Applications

Initial Report

FOOD INGREDIENT RECOGNITION AND RECIPE RECOMMENDATION SYSTEM

(Snap2Cook)

Done by

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Under the guidance of

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2024-2026

MAR ATHANASIUS COLLEGE OF ENGINEERING
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CERTIFICATE

Food Ingredient Recognition and Recipe Recommendation System

Certified that this is the Bonafide record of project work done by

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during the third semester, in partial fulfilment of requirements for award of the degree,

Master of Computer Applications

of

APJ Abdul Kalam Technological University Thiruvananthapuram

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1.INTRODUCTION

Food and nutrition play a vital role in maintaining a healthy lifestyle, and with the rapid advancement of technology, intelligent food-related applications have become increasingly popular. In recent years, the growth of image processing, machine learning, and deep learning has enabled the development of smart systems capable of analyzing visual data with high accuracy. Among these applications, automated food analysis and recommendation systems have gained significant attention due to their practical relevance in daily life.

CNN-based models are capable of automatically learning visual features such as shapes, textures, and patterns directly from images, eliminating the need for manual feature extraction. Transfer learning using pretrained models like ResNet50 allows these systems to achieve high accuracy even with limited training data, making them suitable for food ingredient recognition tasks.

In parallel, recommendation systems have evolved to provide personalized suggestions based on user inputs and data analysis techniques. Content-based filtering is one such approach that recommends items by analyzing their attributes and matching them with user requirements. When applied to food recommendation, content-based filtering techniques such as TF-IDF vectorization and cosine similarity can effectively match recognized ingredients with relevant recipes from large datasets.

This project, titled Snap2Cook, proposes a cloud-based food ingredient recognition and recipe recommendation system that integrates deep learning and content-based recommendation techniques. The system enables users to upload an image of a food ingredient, which is automatically recognized using a CNN-based model. Based on the detected ingredient, suitable recipes are recommended from a large-scale recipe dataset. The system also supports nutrition-aware filtering, such as calorie-based constraints, to promote healthier food choices.

To ensure scalability and real-world usability, the application is developed as a web-based system and deployed on cloud infrastructure. By combining computer vision, text-based recommendation methods, external APIs, and cloud deployment, the proposed system aims to provide an efficient, user-friendly, and practical solution for intelligent food recommendation. This project demonstrates how modern artificial intelligence techniques can be effectively applied to address real-world problems in the domain of food and nutrition.

2.SUPPORTING LITERATURE

2.1 Literature Reviews

Paper 1: Rodrigues MS, Fidalgo F, Oliveira Â. RecipeIS—Recipe recommendation system based on recognition of food ingredients. *Applied Sciences,2023*

Rodrigues et al. proposed a recipe recommendation system named RecipeIS that focuses on recognizing food ingredients from images and recommending recipes accordingly. The primary objective of the study is to assist users in deciding what to cook using the ingredients they already have, thereby reducing food waste and improving decision-making in daily cooking activities. The system emphasizes image-based ingredient recognition rather than traditional text-based recipe search.

The authors employ a Convolutional Neural Network (CNN) using a pretrained ResNet50 model through transfer learning to classify food ingredients. The model is trained on a fruits and vegetables image dataset consisting of 36 ingredient classes. Image preprocessing techniques such as resizing and normalization are applied to improve classification accuracy. The recognized ingredient labels are then used as input to retrieve recipes.

Instead of maintaining a local recipe database, the system integrates the Edamam Recipe API to fetch relevant recipes dynamically. This approach allows the system to provide real-time recipe recommendations without the overhead of managing a large recipe dataset. The experimental results demonstrate that the ResNet50-based CNN achieved high classification accuracy, making it suitable for practical food ingredient recognition tasks.

Overall, the paper highlights the effectiveness of combining deep learning-based ingredient recognition with external recipe APIs to build scalable and user-friendly food recommendation systems. However, the system is limited to fruits and vegetables and depends heavily on third-party API availability.

Title of the paper	RecipeIS—Recipe Recommendation System Based on Recognition of Food Ingredients
Area of work	Image-based food ingredient recognition and recipe recommendation
Dataset	Fruits and Vegetables Image Recognition Dataset (36 ingredient classes)
Methodology/Strategy	Ingredient recognition using CNN (ResNet50) followed by recipe retrieval
Architecture	Pretrained ResNet50 CNN with additional convolutional layers
Recommendation Method	API-based recipe retrieval using Edamam Recipe API
Result/Accuracy	Approximately 96% accuracy for ingredient classification
Advantages	The system enables automatic ingredient recognition from images and provides real-time recipe recommendations using an external API, eliminating the need for maintaining a local recipe database.
Limitations	The system is limited to fruits and vegetables and depends on the availability of the external Edamam Recipe API.
Future Proposals	Includes extending support to multiple ingredients, expanding the dataset, and improving system robustness.

Table 2.1 Summary of Paper 1

Paper 2: Kumar A, Sharma N, Singh R. Food recipe recommendation based on ingredients detection using deep learning. In: *Proceedings of the ACM International Conference on Advances in Computing, Communication and Control*; 2022.

The paper *"Improved Ingredients-Based Recipe Recommendation Software Using Machine Learning"* presented a food recipe recommendation system that detects food ingredients from images and recommends suitable recipes using deep learning techniques. The main objective of the study is to help users identify ingredients visually and suggest recipes that can be prepared using those ingredients, particularly for users who may not know ingredient names.

The authors use a CNN-based deep learning model with transfer learning, comparing multiple pretrained architectures such as VGG16, MobileNet, and ResNet50. Among these models, ResNet50 achieved the highest classification accuracy and was selected as the final model. The CNN is trained on a custom dataset consisting of 9,856 images across 32 food ingredient classes.

For recipe recommendation, the authors designed a structured approach using a two-dimensional matrix that explicitly represents the relationship between ingredients and recipes. Recipes are recommended based on ingredient matching rather than similarity scoring. This rule-based recommendation strategy ensures that suggested recipes contain the recognized ingredients.

Experimental results show that the ResNet50-based model achieved an ingredient classification accuracy of 94%, outperforming other CNN architectures. The study demonstrates the effectiveness of transfer learning for ingredient detection and provides a simple yet effective recipe recommendation mechanism. However, the system supports only single-ingredient detection and lacks advanced personalization features.

Title of the paper	Food Recipe Recommendation Based on Ingredients Detection Using Deep Learning
Area of work	Ingredient detection and recipe recommendation
Dataset	Custom food ingredient image dataset consisting of 32 ingredient classes with a total of 9,856 images collected from various sources.
Methodology/Strategy	CNN with transfer learning
Architecture	CNN architectures including ResNet50, VGG16, and MobileNet, with ResNet50 selected as the best-performing model.
Recommendation Method	A rule-based ingredient–recipe matching approach using a 2D ingredient–recipe matrix to recommend recipes
Result/Accuracy	The ResNet50 model achieved 94% ingredient classification accuracy, outperforming the other CNN models.
Advantages	The approach demonstrates high accuracy in ingredient detection and uses a simple and effective rule-based recipe recommendation strategy.
Limitations	The system supports only single ingredient detection and lacks personalization features.
Future Proposals	The authors suggest extending the system to support multiple ingredients and incorporating advanced recommendation techniques.

Table 2.2 Summary of Paper 2

Paper 3: **R**ahman MA, Hossain MS, Islam R. Improved ingredients-based recipe recommendation software using machine learning. *IEEE Access*. 2024.

The study titled "Improved Ingredients-Based Recipe Recommendation Software Using Machine Learning" presents an advanced recipe recommendation system that integrates deep learning–based ingredient recognition with machine learning–based recommendation techniques. The primary objective of the research is to improve the relevance and personalization of recipe recommendations by combining visual ingredient recognition with user preferences and hybrid filtering strategies.

The authors propose a system that first identifies food ingredients from images using a Convolutional Neural Network (CNN) architecture based on ResNet50V2. Transfer learning is employed to leverage pretrained ImageNet weights, enabling the model to learn discriminative visual features efficiently. The CNN model is trained on a customized ingredient image dataset consisting of 33 food ingredient classes, including fruits, vegetables, meats, and other commonly used ingredients. Once the ingredients are recognized, the output labels are used as input for the recipe recommendation module. The recommendation system adopts a hybrid approach that combines content-based filtering, collaborative filtering, and a weighted average score technique. In the content-based component, recipe ingredient lists are represented using TF-IDF vectorization, and cosine similarity is employed to measure similarity between the recognized ingredients and available recipes. This enables the system to recommend recipes that closely match the detected ingredients.

In addition to content-based filtering, collaborative filtering is incorporated to improve personalization by considering user ratings, preferences, and interaction history. The weighted average score technique is used to rank recipes by combining ingredient similarity scores with user feedback, thereby prioritizing highly relevant and popular recipes. The Food.com dataset is used as the primary source of recipe data due to its large size and rich ingredient and nutritional information.

Experimental evaluation of the proposed system shows that the ResNet50V2-based ingredient recognition model achieves approximately 89–90% validation accuracy, indicating reliable classification performance.

Title of the paper	Improved Ingredients-Based Recipe Recommendation Software Using Machine Learning
Area of work	Ingredient-based personalized recipe recommendation using machine learning and deep learning techniques.
Dataset	Custom ingredient image dataset containing 33 ingredient classes and the Food.com recipe dataset for recommendation.
Methodology/Strategy	The system uses a CNN-based model (ResNet50V2) for ingredient classification. Recipe recommendation is performed
Architecture	ResNet50V2 CNN architecture integrated with hybrid recommendation modules.
Recommendation Method	It performed using a hybrid approach combining content-based filtering, collaborative filtering, and a weighted average scoring technique.
Result/Accuracy	The ingredient classification model achieved approximately 89–90% validation accuracy, and the hybrid recommendation approach improved recommendation relevance.
Advantages	The system provides personalized recipe recommendations by combining ingredient recognition with user preferences and multiple filtering techniques.
Limitations	The hybrid recommendation approach increases system complexity and computational overhead.
Future Proposals	Future enhancements include improving scalability, expanding datasets, and optimizing recommendation efficiency.

Table 2.3 Summary of Paper 3

2.2 Literature Review Summary

Paper	Summary
Paper 1: Rodrigues MS, Fidalgo F, Oliveira Â. RecipeIS—Recipe recommendation system based on recognition of food ingredients. *Applied Sciences,2023*	This paper presents an image-based recipe recommendation system that identifies food ingredients using a CNN with a pretrained ResNet50 model. The system retrieves recipes dynamically using the Edamam Recipe API instead of maintaining a local recipe database. Experimental results show high ingredient recognition accuracy, demonstrating the effectiveness of deep learning for practical food recommendation applications.
Paper 2: Kumar A, Sharma N, Singh R. Food recipe recommendation based on ingredients detection using deep learning. In: *Proceedings of the ACM International Conference on Advances in Computing, Communication and Control*; 2022.	This study focuses on detecting food ingredients from images using CNN models and recommending recipes based on ingredient matching. Multiple pretrained models were evaluated, with ResNet50 achieving the highest accuracy of 94%. The paper demonstrates the effectiveness of transfer learning for ingredient classification but uses a simple rule-based recommendation approach.
Paper 3: **R**ahman MA, Hossain MS, Islam R. Improved ingredients-based recipe recommendation software using machine learning. *IEEE Access*. 2024.	This paper proposes a hybrid recipe recommendation system that combines CNN-based ingredient recognition with content-based and collaborative filtering techniques. A ResNet50V2 model is used for ingredient classification, and recipe relevance is improved using TF-IDF, cosine similarity, and user preferences. The hybrid approach enhances recommendation accuracy but increases system complexity.

Table 2.4 Overall Summary of all Papers

2.3 Findings and Proposals

The review of existing literature reveals that recent advancements in deep learning, particularly Convolutional Neural Networks (CNNs), have significantly improved the accuracy of food ingredient recognition from images. Multiple studies demonstrate that pretrained CNN architectures such as ResNet50 and ResNet50V2, when used with transfer learning, outperform traditional image processing and machine learning techniques. These models are effective even with limited datasets, making them suitable for real-world food recognition applications.

The literature further indicates that ingredient-based recipe recommendation is an effective approach for assisting users in deciding what to cook using available ingredients. Content-based filtering techniques, especially TF-IDF vectorization and cosine similarity, are widely used to match recognized ingredients with relevant recipes from large-scale datasets such as Food.com. Some studies enhance recommendation relevance by integrating collaborative filtering and user preferences; however, such hybrid approaches increase system complexity. Additionally, the use of external recipe APIs has been shown to improve recommendation coverage and provide real-time results, though complete dependency on APIs can limit system control and scalability.

Based on these findings, this project proposes the development of an intelligent food ingredient recognition and recipe recommendation system. The proposed system will employ a CNN-based deep learning model using a pretrained ResNet50 architecture to automatically identify food ingredients from user-uploaded images. For recipe recommendation, a content-based filtering approach will be applied to the Food.com dataset using TF-IDF and cosine similarity to ensure relevant and accurate recipe matching. To promote healthier choices, the system will support nutrition-aware filtering, such as calorie-based constraints. An external recipe API will be optionally used as a supplementary source to enhance recommendation coverage.

The proposed system will be implemented as a web-based application and deployed on a cloud platform to ensure scalability and real-world usability. Overall, the project aims to integrate deep learning, text-based recommendation techniques, and cloud deployment into a practical and efficient food recommendation solution.

3. SYSTEM ANALYSIS

3.1 Analysis of Dataset

3.1.1 About the Dataset

The datasets used in this project are collected from Kaggle.com, an online platform and community for data scientists and machine learning practitioners. Kaggle is a subsidiary of Google LLC and provides access to publicly available datasets, enabling users to explore data, build models, and perform analysis in a web-based environment.

A dataset is a structured collection of data used for training, testing, and validating machine learning and deep learning models. In image-based and recommendation-based systems, the performance of the model highly depends on the quality, size, and diversity of the dataset. Properly prepared and preprocessed datasets are essential for achieving accurate and reliable results.

This project utilizes two datasets:

1. A food ingredient image dataset for ingredient recognition
2. A recipe dataset for content-based recipe recommendation

Dataset 1: Fruits and Vegetables Image Recognition Dataset

Source of the Dataset**:**
https://www.kaggle.com/datasets/kritikseth/fruit-and-vegetable-image-recognition

The Fruits and Vegetables Image Recognition Dataset is a publicly available dataset designed for food ingredient classification tasks. It contains labeled images of commonly used fruits and vegetables collected from various online sources.

The dataset consists of images belonging to 36 different food ingredient classes, including both fruits and vegetables. Each class contains multiple images, resulting in approximately 3,600 images in total. The images are collected from various online sources and are labeled according to their corresponding ingredient category.

The images vary in resolution, lighting conditions, background, and orientation. This variation makes the dataset suitable for training convolutional neural network (CNN) models, as it helps the model learn robust and generalized visual features applicable to real-world scenarios.

This dataset is used for training and evaluating the deep learning model for automatic food ingredient recognition from images**.**

Dataset 2: Food.com Recipe Dataset

Source of the Dataset**:**

Food.com https://www.kaggle.com/datasets/irkaal/foodcom-recipes-and-reviews

The Food.com dataset is a large-scale recipe dataset containing detailed information about recipes collected from the Food.com platform. The dataset is provided in CSV format and contains approximately 28 different attributes related to recipes.

The dataset includes lakhs of recipe records, making it suitable for large-scale recipe analysis and recommendation tasks. Each recipe entry contains information such as recipe name, ingredient list, cooking instructions, calorie values, and user ratings.

This dataset is used for implementing content-based recipe recommendation by matching recognized food ingredients with relevant recipes.

3.1.2 Explore the Dataset

The Fruits and Vegetables Image Recognition Dataset is organized into three main folders: train, validation, and test, each containing the same 36 food ingredient classes. The classes include a wide range of fruits and vegetables such as apple, banana, tomato, onion, potato, carrot, cabbage, garlic, mango, grapes, and watermelon. This consistent class distribution across all splits ensures balanced training and evaluation of the model. The dataset images have varying resolutions, with common image sizes including 1000×1000, 800×800, 1024×768, 600×600, and 1200×800, reflecting real-world variability in image capture. The presence of diverse image dimensions, backgrounds, and lighting conditions enhances the robustness and generalization capability of the deep learning model.

```
print("Folders:", os.listdir(dataset_path))
train_dir = os.path.join(dataset_path, "train")
val_dir   = os.path.join(dataset_path, "validation")
test_dir  = os.path.join(dataset_path, "test")
print("Dataset root folders:", os.listdir(dataset_path))

for split in ["train", "validation", "test"]:
    split_dir = os.path.join(dataset_path, split)
    print(f"\n{split.upper()} classes ({len(os.listdir(split_dir))}):")
    print(sorted(os.listdir(split_dir)))
```

```
Folders: ['validation', 'train', 'test']
Dataset root folders: ['validation', 'train', 'test']

TRAIN classes (36):
['apple', 'banana', 'beetroot', 'bell pepper', 'cabbage', 'capsicum', 'carr

VALIDATION classes (36):
['apple', 'banana', 'beetroot', 'bell pepper', 'cabbage', 'capsicum', 'carr

TEST classes (36):
['apple', 'banana', 'beetroot', 'bell pepper', 'cabbage', 'capsicum', 'carr
```

The Food.com recipe dataset is a large-scale collection of recipe data provided in CSV format, containing approximately 28 different attributes for each recipe. The dataset consists of lakhs of recipe records, making it suitable for large-scale recipe analysis and recommendation tasks.

Each record in the dataset represents a unique recipe and includes detailed information such as the recipe name, list of ingredients, cooking instructions, calorie content, and user ratings. Among the available attributes, only the relevant features required for recipe recommendation are selected for this project.

Before using the dataset for recommendation, preprocessing steps such as removal of missing values, text normalization, and ingredient list cleaning are performed. The ingredient text is then converted into a suitable format for applying TF-IDF vectorization and cosine similarity.

3.2 Data Preprocessing

3.2.1 Data Cleaning

The Fruits and Vegetables Image Recognition Dataset

a. Removal of Irrelevant and Inconsistent Classes

Initially, the dataset contained some classes that were either inconsistent in naming or not required for the scope of the project. These classes were removed from all dataset splits (training, validation, and testing) to maintain uniformity.

The following classes were removed: Sweetcorn, Paprika, Jalepeno, Bell pepper. This removal was performed across the train, validation, and test directories to ensure consistency in class distribution

```
import shutil
classes_to_remove = ["sweetcorn","paprika","jalepeno","bell pepper"]

def remove_classes(base_dir, classes):
    for cls in classes:
        cls_path = os.path.join(base_dir, cls)
        if os.path.exists(cls_path):
            shutil.rmtree(cls_path)
            print(f"Removed: {cls_path}")

for split in ["train", "validation", "test"]:
    remove_classes(os.path.join(dataset_path, split), classes_to_remove)
```

```
th/fruit-and-vegetable-image-recognition/versions/8/train/sweetcorn
th/fruit-and-vegetable-image-recognition/versions/8/train/paprika
th/fruit-and-vegetable-image-recognition/versions/8/train/jalepeno
th/fruit-and-vegetable-image-recognition/versions/8/train/bell pepper
th/fruit-and-vegetable-image-recognition/versions/8/validation/sweetcorn
th/fruit-and-vegetable-image-recognition/versions/8/validation/paprika
th/fruit-and-vegetable-image-recognition/versions/8/validation/jalepeno
th/fruit-and-vegetable-image-recognition/versions/8/validation/bell pepper
th/fruit-and-vegetable-image-recognition/versions/8/test/sweetcorn
th/fruit-and-vegetable-image-recognition/versions/8/test/paprika
```

b. Image Resizing & Augmentation

In this project, all food images are resized to 224 × 224 pixels with 3 color channels (RGB**)**. This size is chosen because it is the required input dimension for the ResNet50 pretrained model. Resizing ensures consistency across the dataset and enables efficient batch processing during training and inference. Image resizing and data augmentation are performed using ImageDataGenerator. Images are resized to 224×224 pixels to match ResNet50 input requirements, and appropriate preprocessing and augmentation techniques are applied to improve model generalization.

```
IMG_SIZE = (224, 224)
train_datagen = ImageDataGenerator(
    preprocessing_function=preprocess_input,
    rotation_range=30,
    width_shift_range=0.1,
    height_shift_range=0.1,
    shear_range=0.1,
    zoom_range=0.2,
    horizontal_flip=True,
    fill_mode="nearest"
)

val_datagen = ImageDataGenerator(
    preprocessing_function=preprocess_input
)
```

Food.com recipe dataset

a. Selection of Relevant Ingredients

To align the recipe dataset with the food ingredient classification model, a predefined list **of** target ingredients (fruits and vegetables used in the CNN model) was created. Only recipes containing at least one of these target ingredients were retained.

b. Feature Selection

From the original dataset, only the relevant columns required for recommendation were selected:

- Recipe ID, Recipe Name, Ingredient Parts, Ingredient Quantities, Recipe Instructions
- Calories, Protein Content, Fat Content, Carbohydrate Content,images

This reduced data redundancy and improved processing efficiency.

c. Duplicate and Missing Value Handling

Duplicate recipes were removed using the unique RecipeId.
Records with missing values in critical fields such as ingredient parts, quantities, instructions, and calories were discarded.For nutritional attributes (fat, protein, and carbohydrate content), missing values were replaced with zero to maintain numerical consistency.

d. Ingredient Parsing and Normalization

Ingredient parts and quantities were parsed into structured lists using a custom parser. All ingredient names were converted to lowercase to ensure consistency.

To reduce noise and improve similarity matching, common non-informative words such as fresh, chopped, cloves, oil, large, and small were removed from ingredient names. This normalization step improves the effectiveness of TF-IDF vectorization.

e. Combining Ingredients with Quantities

Ingredient quantities and ingredient names were combined into a single structured representation. If quantity information was missing, only the ingredient name was retained. This step preserves useful cooking information while maintaining textual consistency.

3.2.2 Analysis of Feature

Features refer to the input attributes used by the model to learn patterns and make predictions. In this project, features are derived from image data and recipe data, depending on the stage of processing.

In this project, the Fruits and Vegetables Image Recognition Dataset is used for food ingredient classification. The dataset consists of images belonging to 32 food ingredient categories after preprocessing. Each category represents a distinct fruit or vegetable and is stored in a separate folder, which helps in supervised learning and class-wise labeling.

All images are resized to a uniform dimension of 224 × 224 pixels with 3 color channels (RGB) to meet the input requirements of the ResNet50 model. The pixel values of these images serve as the primary features for the model. These features capture important visual characteristics such as color, texture, edges, and shape of food ingredients.

For the recipe recommendation module, features are derived from the ingredient lists in the Food.com dataset. The ingredient text is preprocessed and transformed into numerical feature vectors using TF-IDF (Term Frequency–Inverse Document Frequency)**.** These vectors represent the importance of each ingredient in a recipe and enable similarity comparison between recognized ingredients and available recipes

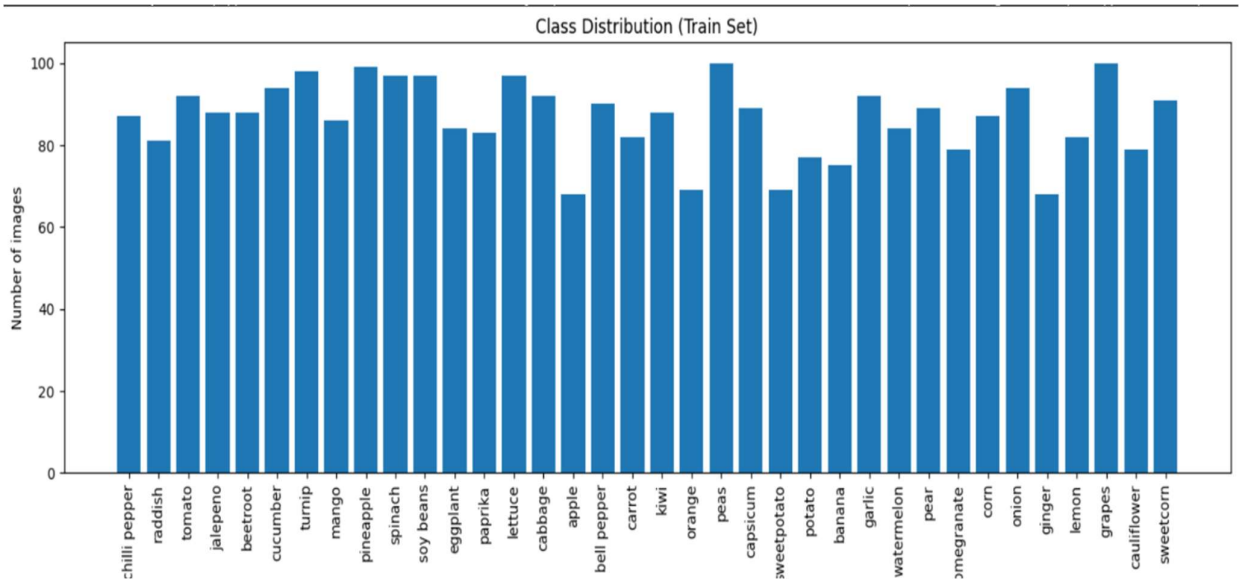
3.2.3 Analysis of Class Variables

Class variables represent the target output labels that the model is trained to predict. In this project, the class variables correspond to the **f**ood ingredient categories present in the image dataset.

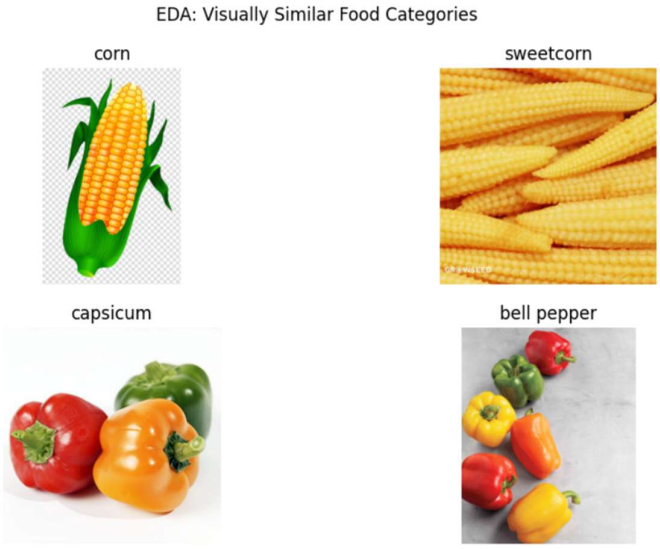
After preprocessing and cleaning, the dataset contains 32 distinct food ingredient classes, including fruits and vegetables such as apple, banana, tomato, onion, potato, carrot, cabbage, garlic, mango, watermelon, and others. Each class represents a unique food ingredient that the CNN model is trained to recognize.

These class variables are categorical in nature and are encoded using one-hot encoding for training the CNN model. During training, the model learns to map extracted image features to one of the predefined ingredient classes.

3.3 Data Visualization



Class Distribution before preprocessing



Visually Similar Food Categories

3.4 Analysis of Architecture

3.4.1 Detailed study of Architecture

The proposed system uses a Convolutional Neural Network (CNN) based on the ResNet50 architecture for automatic food ingredient recognition from images. CNNs are a class of deep neural networks specifically designed to process visual data by learning spatial hierarchies of features directly from input images.

In this project, transfer learning is employed by using a pretrained ResNet50 model trained on the ImageNet dataset. Instead of training the network from scratch, the pretrained model is used as a feature extractor, and additional layers are added to adapt the network to the food ingredient classification task.

ResNet50 is a deep residual network consisting of 50 layers, designed to overcome the vanishing gradient problem by introducing residual (skip) connections. These connections allow the network to learn identity mappings and enable effective training of very deep networks.

Characteristics and Features of ResNet50-based CNN

- **Deep Architecture:**
  ResNet50 consists of 50 layers, allowing the model to learn complex and high-level visual features from food image.
- **Residual Connections:**
  Skip connections enable gradients to flow directly through the network, preventing performance degradation in deep models.
- **Transfer Learning:**
  The pretrained weights from ImageNet are reused, reducing training time and improving accuracy with limited data.
- **Frozen Base Layers:**
  All layers of the ResNet50 base model are frozen during training to preserve learned visual features and reduce overfitting.
- **Custom Classification Head:**
  Additional dense layers are added to adapt the model to the specific number of food ingredient classes.

Layer-wise Description of the Proposed Model

Input Layer: The input layer accepts food images resized to 224 × 224 × 3, where 3 represents the RGB color channels. No computation is performed at this stage**.**

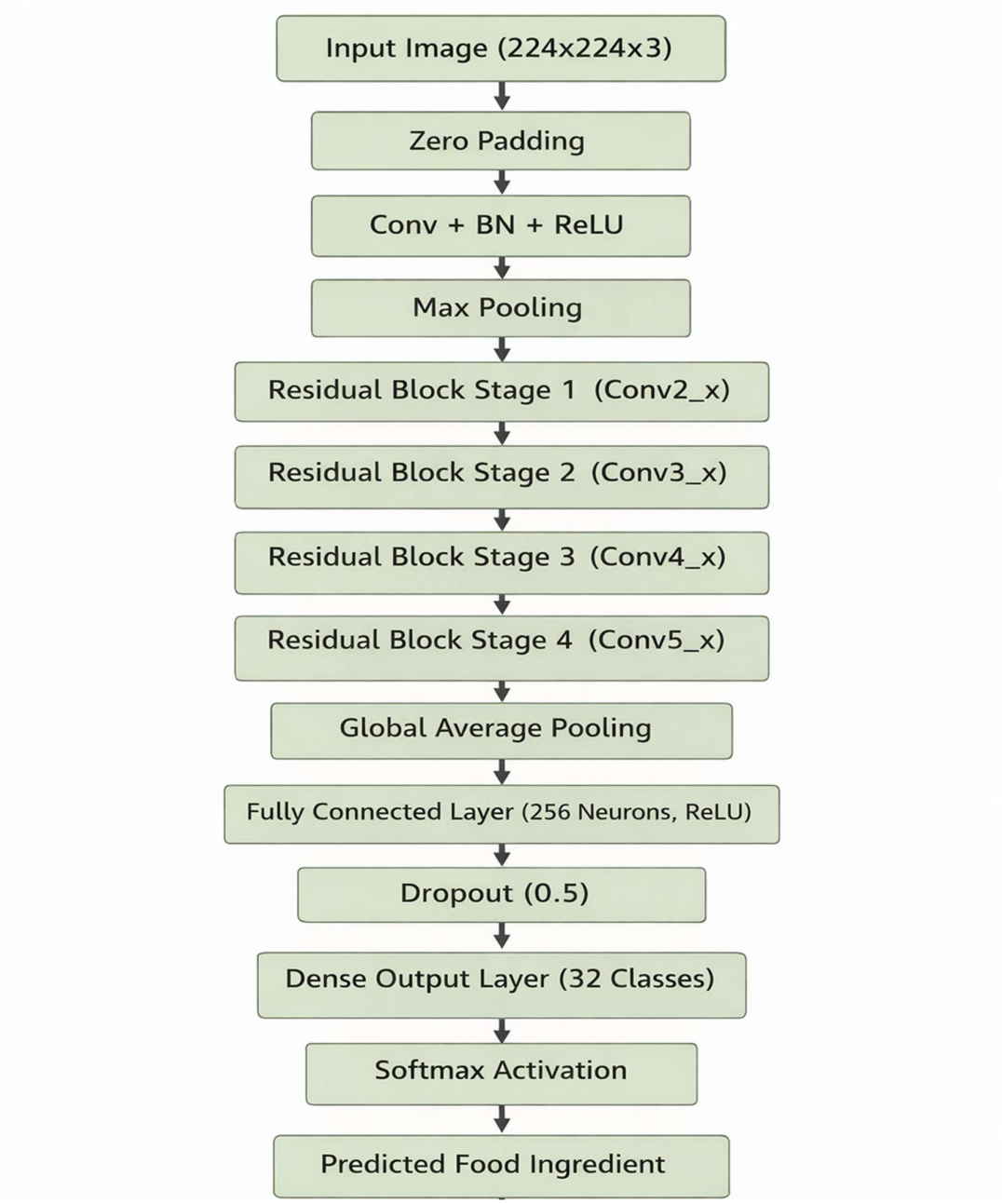
ResNet50 Convolutional Layers: The ResNet50 base model consists of multiple convolutional layers organized into residual blocks. These layers extract low-level features such as edges and textures in early layers and high-level features such as shapes and patterns in deeper layers. All ResNet50 layers are frozen during training.

Global Average Pooling Layer: After feature extraction, a Global Average Pooling layer is applied to reduce the spatial dimensions of the feature maps. This layer converts the feature maps into a single feature vector and reduces the number of parameters, helping to prevent overfitting.

Fully Connected Dense Layer: A dense layer with 256 neurons and ReLU activation is added to learn task-specific representations relevant to food ingredient classification.

Dropout Layer: A dropout layer with a dropout rate of 0.5 is applied to randomly disable neurons during training. This improves generalization and reduces overfitting.

Output Layer: The final dense layer contains neurons equal to the number of food ingredient classes. A Softmax activation function is applied to produce probability scores for each class. The class with the highest probability is selected as the predicted ingred



Block diagram of architecture

3.4.2 Recommendation Model Technique – Content-Based Filtering

Ingredient Representation Using TF-IDF

To represent recipe ingredients in a meaningful numerical form, the Term Frequency–Inverse Document Frequency (TF-IDF) technique is employed. TF-IDF assigns a weight to each ingredient based on:

- How frequently the ingredient appears in a recipe (term frequency)
- How rare the ingredient is across all recipes (inverse document frequency)

The TF-IDF weight is computed as:

$$w_{i,j} = tf_{i,j} \times \log\left(\frac{N}{df_i}\right)$$

Where:

- tfi,j is the frequency of ingredient *i* in recipe *j*
- dfi is the number of recipes containing ingredient *i*
- N is the total number of recipes

This representation helps emphasize important ingredients while reducing the influence of common ones.

Similarity Measurement Using Cosine Similarity

After TF-IDF vectorization, cosine similarity is used to measure the similarity between the predicted ingredient vector and recipe ingredient vectors. Cosine similarity is well-suited for sparse, high-dimensional data such as text features.

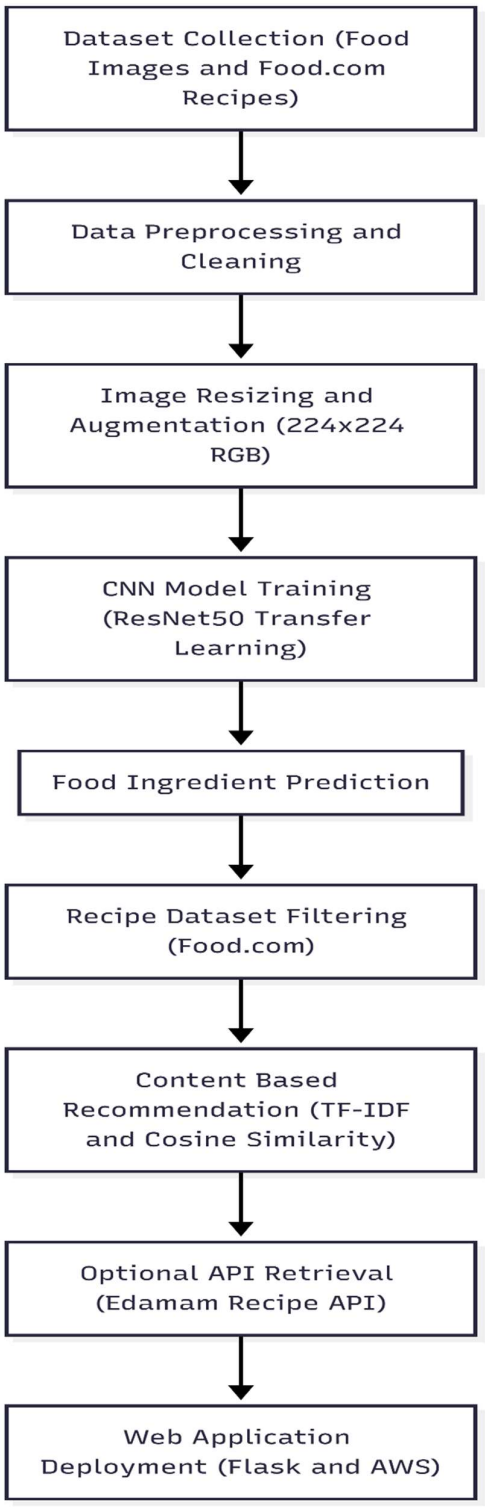
The cosine similarity between two vectors A and B is defined as:

$$\cos(\theta) = \frac{A.B}{||A||.||B||} = \frac{\sum_i^n A_i B_i}{\sqrt{\sum_i^n A_i^2}\sqrt{\sum_i^n B_i^2}}$$

A higher cosine similarity score indicates a stronger match between the recognized ingredient and the recipe's ingredient list.

3.5 Project Pipeline

3.5.1 Project pipeline diagram



Project Pipeline

Explanation of Each Pipeline Level

1. Dataset Collection

Food images are collected from the Fruits and Vegetables Image Recognition Dataset (Kaggle), and recipe data is obtained from the Food.com dataset. These datasets form the foundation of the system.

2. Data Preprocessing & Cleaning

Unwanted classes are removed, duplicate images are detected and eliminated, and class consistency is ensured. Recipe data is cleaned by handling missing values, normalizing ingredient text, and removing duplicates. All images are resized to 224 × 224 × 3 to match ResNet50 input requirements. Data augmentation techniques such as rotation, zooming, and flipping are applied to improve model generalization.

3. CNN Model Training (ResNet50)

A pretrained ResNet50 model is used with transfer learning. The base model is frozen, and a custom classifier is trained to classify food ingredient images into predefined classes.

4. Food Ingredient Prediction

When a user uploads an image, the trained CNN model predicts the corresponding food ingredient label.

5. Recipe Dataset Filtering

Recipes containing the predicted ingredient are retrieved from the preprocessed Food.com dataset to reduce search space and improve recommendation relevance.

6. Content-Based Recommendation

TF-IDF vectorization is applied to recipe ingredient lists, and cosine similarity is used to rank recipes based on similarity to the predicted ingredient.

7. Optional API-Based Recipe Retrieval

If insufficient recipes are available locally, the Edamam Recipe API is used as a fallback to fetch additional real-time recipe suggestions.

8. Web Application Deployment

The complete system is integrated into a Flask-based web application and deployed using AWS services for real-world accessibility.

Project Development Plan (Phase-wise)

Phase 1: Requirement Analysis

- Study of existing food recommendation systems
- Identification of system objectives and scope
- Selection of datasets and technologies

Phase 2: Dataset Collection & Analysis

- Downloading image and recipe datasets
- Understanding dataset structure and attributes
- Exploratory data analysis

Phase 3: Data Preprocessing

- Image cleaning, class removal, and duplicate handling
- Recipe dataset filtering and normalization
- Feature extraction preparation

Phase 4: Model Design & Training

- Selection of ResNet50 architecture
- Transfer learning implementation
- Model training and validation

Phase 5: Recommendation System Development

- Implementation of TF-IDF vectorization
- Cosine similarity-based ranking
- Nutrition-aware filtering

Phase 6: Web Application Development

- Frontend design using HTML, CSS, Bootstrap
- Backend integration using Flask
- Model inference and recipe retrieval integration

Phase 7: Deployment & Testing

- Deployment on AWS EC2 and S3
- Functional testing and performance evaluation
- System validation

3.6 Feasibility Analysis

A feasibility study aims to objectively evaluate the strengths and weaknesses of the proposed system, identify opportunities and constraints, and determine whether the project can be successfully implemented using available resources. The feasibility of the proposed Food Ingredient Recognition and Recipe Recommendation System is evaluated under the following categories:

- Technical Feasibility
- Economic Feasibility
- Operational Feasibility

3.6.1 Technical Feasibility

The proposed system is technically feasible, as it is developed using well-established and widely available technologies. Recent advancements in deep learning and computer vision have demonstrated high accuracy in food image recognition tasks using Convolutional Neural Networks (CNNs).

The system employs a pretrained ResNet50 model, which has proven effectiveness in image classification tasks. The availability of publicly accessible datasets such as the Fruits and Vegetables Image Recognition Dataset and the Food.com recipe dataset further supports feasibility.
The integration of content-based filtering techniques using TF-IDF and cosine similarity for recipe recommendation is a standard and reliable approach. Additionally, the use of Flask for web development and AWS cloud services for deployment ensures scalability and reliability.

Hence, the proposed system is technically feasible with the current tools and technologies.

3.6.2 Economic Feasibility

The proposed system is economically feasible, as it requires minimal financial investment. The development is carried out using open-source software and libraries, such as Python, TensorFlow, Keras, and Scikit-learn.

Model training and experimentation are performed using Google Colab and Kaggle notebooks, which provide free access to GPU resources. This eliminates the need for expensive hardware during development.

Deployment on AWS EC2 and S3 can be carried out using low-cost or free-tier services, making the overall cost manageable. Therefore, the system can be developed and deployed without incurring significant expenses.

3.6.3 Operational Feasibility

The proposed system is operationally feasible, as it is designed to be user-friendly and easy to operate. Users can upload food images through a web interface and receive recipe recommendations without requiring technical knowledge.

The backend processes, including image preprocessing, ingredient prediction, and recipe retrieval, are automated. The system can be accessed through any device with an internet connection, making it suitable for real-world usage.

Cloud deployment ensures reliable operation, scalability, and ease of maintenance. Hence, the system is operationally feasible for both developers and end users.

3.7 System Environment

The system environment specifies the hardware and software requirements necessary for the development and execution of the proposed system. A well-defined system environment ensures smooth development, deployment, and maintenance of the application.

3.7.1 Software Environment

The software tools and technologies used in the development of the proposed system are listed below:

1. Python

Python is a high-level programming language used for implementing deep learning models and backend logic. It provides extensive libraries for data processing, machine learning, and web development.

2. TensorFlow and Keras

TensorFlow is an open-source deep learning framework developed by Google. Keras is a high-level API built on TensorFlow that simplifies the creation and training of neural networks. These libraries are used to build and train the CNN model.

3. NumPy and Pandas

NumPy is used for numerical computations and array handling, while Pandas is used for data manipulation and preprocessing of the Food.com recipe dataset.

4. Scikit-learn

Scikit-learn is used for implementing TF-IDF vectorization and cosine similarity, which form the core of the content-based recipe recommendation system.

5. Flask

Flask is a lightweight Python web framework used to develop the web application and integrate the trained model with the user interface.

6. Google Colab

Google Colab provides a cloud-based environment with GPU support for training and testing deep learning models.

7. Kaggle

Kaggle notebooks are used for dataset handling and model experimentation, offering high computational power and easy dataset access.

8. Visual Studio Code

Visual Studio Code is used as the primary code editor for developing the web application and managing project files.

9. HTML, CSS, and Bootstrap

These technologies are used to design and style the user interface, making the web application interactive and user-friendly.

10. GitHub

GitHub is used for version control and project collaboration, enabling code management and tracking of changes.

3.7.2 Hardware Environment

The hardware requirements for developing and running the proposed system are minimal and easily accessible:

- **Processor:** Intel Core i5 or higher
- **Memory:** 8 GB RAM or higher
- **Storage:** 40 GB or more